# In silico identification and characterization of peptidyl-prolyl *cis-trans* isomerases in *Penicillium* and their expression analysis in a halotolerant *P. oxalicum*


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## Introduction

Peptidyl prolyl *cis-trans* isomerases (PPIases), that catalyze *cis-trans* isomerization of the peptidyl-prolyl bond are ubiquitous proteins present in organisms ranging from archaea and bacteria to plants, fungi and animals. Recent studies suggest that though conserved, these proteins are highly versatile and have been implicated in diverse functions such as in maturation of proteins and their trafficking, apoptosis, receptor signaling, immune response, RNA processing, signal transduction, chaperonic activity, etc. The PPIases comprise of different families viz., cyclophilins, FK506-binding proteins (FKBPs), parvulins and protein phosphatase 2A phosphatase activators (PTPAs). Though PPIase gene families have been characterized in different organisms, information regarding these proteins is lacking in *Penicillium* species, which are commercially an important fungi group. In this study, we carried out genome-wide analysis of PPIases in different *Penicillium* spp. and investigated their regulation by salt stress in a halotolerant strain of *Penicillium oxalicum*.

## Materials and methods

### a) In Silico:

The amino acid sequences of cyclophilins, FKBPs, parvulins and PTPAs of 24 different *Penicillium* spp. were retrieved from the NCBI server on the basis of the results obtained with a series of iterative BLASTP homology searches using a variety of PPIases viz., cyclophilin A (hCYPA), FKBP12 (hFKBP12) and parvulins (hPAR14/hPIN1) from human, and PTPAs (YPA1/YPA2) from *Saccharomyces cerevisiae* as queries. The domains in these putative PPIase proteins were identified with PFAM and Prosite database servers using default search parameters. Subcellular localization were predicted with LocTree3, while the presence of Nuclear Localization Signals, Signal peptides and transmembrane domains was further confirmed with NLS mapper, SignalP4.0 and TMHMM ServerNv2.0 servers, respectively, using their default setups. The predicted molecular weights and the isoelectric points were determined by the compute_pi server. Multiple sequence alignments of amino acid sequences belonging to PPIases were performed using MUSCLE algorithm in Jalview standalone package version 2.11.1.3 with default parameters. The Jpred3 server integrated in Jalview was used to predict secondary structure elements. The phylogenetic tree was constructed on the Mega X platform by generating a multiple sequence alignment using the built in MUSCLE algorithm and then employing neighbor-joining (NJ) method with bootstrap value of 1000. The tree files thus obtained were visualized and annotated using Iterative Tree of Life. The MEME web server was used to analyze conserved and potential motifs. Exon–intron organization was predicted using the Gene Structure Display Server 2.0 (GSDS 2.0) by comparing the coding sequences with corresponding DNA sequences.

### b) Wet lab:

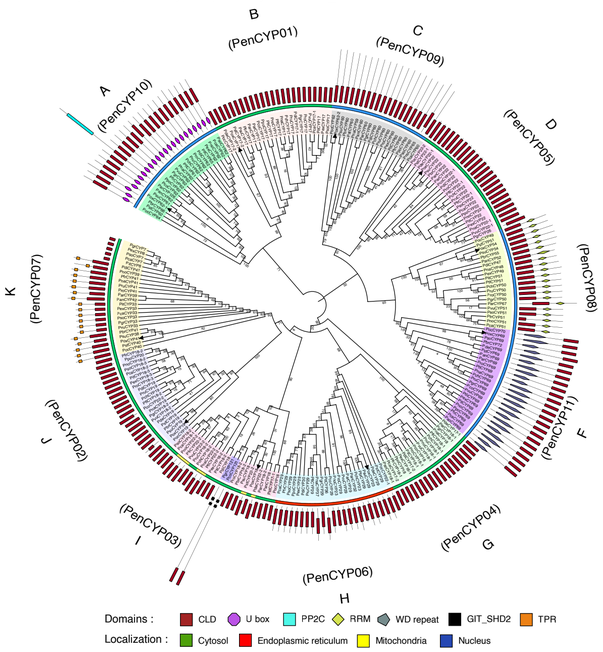
The halotolerant *P. oxalicum* HP1 fungal strain used in this study was grown in Sabouraud production medium, that either lacked or contained NaCl (15%), at 30 °C with shaking at 180 rpm in orbital shaker. The cultures at 4, 7 and 10 days after inoculation (DAI) were taken for total protein extraction and RNA preparation. The total soluble proteins were extracted from the filtered mycelia of *P. oxalicum* by using lysis buffer [50 mm Tris–Cl , 150 mm NaCl (pH 8), 1 mm EDTA (pH 8), 10% Glycerol, 1 mm phenylmethylsulfonyl fluoride]. The total soluble proteins were estimated according to Bradford's method (Bradford 1976) and PPIase activity in the crude extracts was estimated by a chymotrypsin-based coupled reaction at 15 °C for 360 s at 390 nm by using Spectrophotometer. The cyclophilin- and FKBP-associated PPIase activities were determined by the extent of inhibition of reaction in the presence of the specific inhibitors cyclosporin A (CsA) and FK506, respectively. Total RNA from the harvested mycelia was isolated at different stages of growth using a Trizol reagent. After removing DNA by DNaseI (Sigma-Aldrich) treatment, the RNA was quantified, and its integrity was confirmed by denaturing agarose gel electrophoresis (1.4%) followed by staining with ethidium bromide. Superscript III Firststrand synthesis system kit (Invitrogen) was used to synthesize cDNA from 5 µg RNA using Random Hexamer primers.The expression of different cyclophilin, FKBP, parvulin and PTPA genes in *P. oxalicum* was analyzed by the real-time PCR using gene-specific primers.

## Results

In silico analyses revealed that the number of genes encoding cyclophilins, FKBPs, parvulins and PTPAs in *Penicillium* spp. varies between 7-11, 2-5, 1-2, and 1-2, respectively.

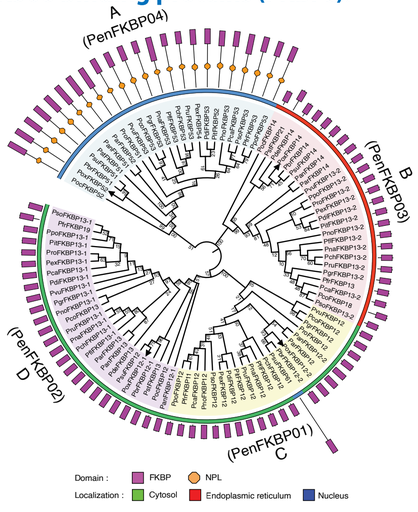
## Cyclophilins:

This study resulted in identification of 237 full-length CLD-containing putative cyclophilin proteins in different species of *Penicillium*. The number of cyclophilins in different *Penicillium* spp. ranges between 7–11, with the *P. oxalicum* genome encoding 10 cyclophilins. Based on homology, the *Penicillium* cyclophilins were clustered into 12 different orthogroups viz., PenCYP01-PenCYP12. Though predominantly cytosolic, the cyclophilins in *Penicillium* were also predicted to localize to the nucleus, endoplasmic reticulum (ER) and mitochondria, highlighting functional divergence of these proteins. Based on the presence of domains other than CLD, the cyclophilins were further classified as single domain (SD) or multidomain (MD) proteins. While seven orthogroups (PenCYP01-PenCYP06 and PenCYP09) consist of SD cyclophilins, five orthogroups (PenCYP07, PenCYP08, PenCYP10-PenCYP12) comprise of MD proteins that contain additional domains such as TPR, RRM, U-box, WD, PP2C, and GIT_SDH.




The phylogenetic relationship of different Penicillium cyclophilins. The unrooted tree was generated using the neighbor-joining method in MEGA X (v10.1.7) software. The constructed tree was further annotated with the Iterative Tree of Life and redrawn in Adobe Illustrator (v25.2.3). Bootstrap values from 1000 replicates are indicated at each branch. The localization of each member is represented in different colors. Domain architecture of each protein is also shown in the outermost layer. Black triangles represent the *Penicillium oxalicum*.

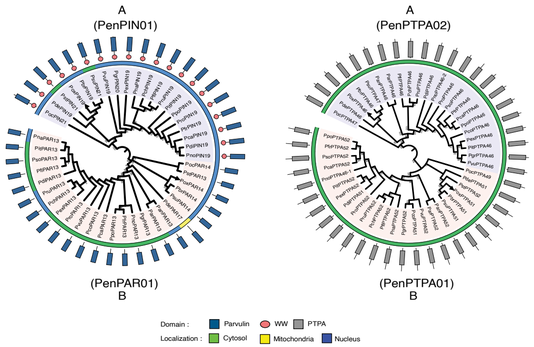
## FK506 binding proteins (FKBPs):




Ninety-three putative FKBPs were identified in different *Penicillium* species. Based on similarity, these proteins were categorized into four different orthogroups viz., PenFKBP01, PenFKBP02, PenFKBP03 and PenFKBP04. This grouping was also supported by the phylogenetic analysis, which depicted a close relationship of these proteins within a group. The FKBPs in *Penicillium* spp. were predicted to localize to different subcellular organelles. While members of the orthogroups PenFKBP01 and PenFKBP02 may localize to the cytosol, the PenFKBP03 and PenFKBP04 FKBPs are likely to be present in the ER and nucleus, respectively. Contrary to the PenFKBP01, PenFKBP02 and PenFKBP03 proteins, which consist of only FKBP domain, the PenFKBP04 members also exhibited a nucleoplasmin like (NPL) domain.

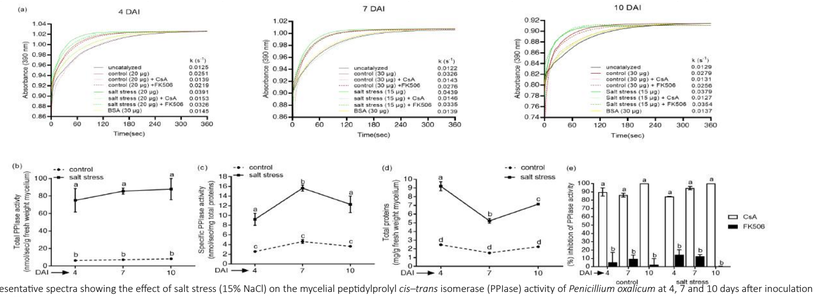
## Parvulins and PTPAs:

The *Penicillium* parvulins were grouped into two different orthogroups viz., PenPIN01 and PenPAR01, respectively. Parvulins were predicted to localize to cytosol, mitochondria and nucleus. Contrary to the PenPAR01 parvulins, that contain only the PPIase domain, the PenPIN01 members also showed an additional conserved N-terminal WW domain. Two different orthogroups viz., PenPTPA01 and PenPTPA02 were identified in PTPAs and were localized in cytosol.




## PPIase activity:

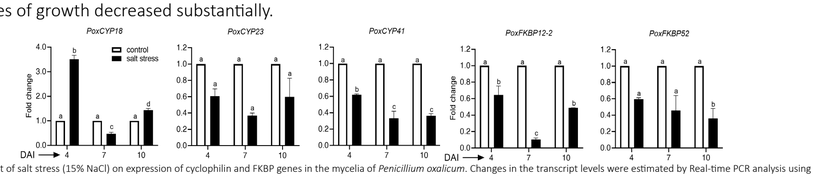
The total (nmol/s/g fresh weight mycelium) and specific (nmol/s/mg total proteins) PPIase activities under salt stress were significantly higher than control at all the three stages of growth in *P. oxalicum*. PPIase activity was almost completely inhibited by CsA at all the growth stages, as compare to FK506 inhibitor. These observations imply that PPIase activity in the mycelia of *P. oxalicum* was predominantly contributed by the cyclophilins.




(a) Representative spectra showing the effect of salt stress (15% NaCl) on the mycelial peptidylprolyl *cis–trans* isomerase (PPIase) activity of *Penicillium oxalicum* at 4, 7 and 10 days after inoculation (DAI). The PPIase activity in the crude mycelial extracts was estimated by studying the rate of chymotrypsin catalyzed cleavage of the test peptide, and expressed as the first-order rate constant, k (s–1). Higher first-order rate constant signified greater PPIase activity in the extract. 20 µg of total proteins was used for this assay, with bovine serum albumin (BSA) as a negative control. The cyclophilin- and FK506-binding protein (FKBP)-associated PPIase activity was estimated by the extent of inhibition in the presence of 100 nM cyclosporin A (CsA) and 2 µM FK506, respectively. Control refers to the cultures grown in Sabouraud medium without supplementing with NaCl. (b) Changes in the total PPIase activity (nmol/sec/g fresh weight mycelium) and (c) specific PPIase activity (nmol/sec/mg total proteins), and (d) total soluble proteins (mg/g fresh weight mycelium) in the mycelia of Penicillium oxalicum grown in the presence or absence of salt. (e) Effect of specific inhibitors of cyclophilins (CsA) and FKBPs (FK506) on mycelial PPIase activity. The percent inhibition of PPIase activity is expressed with respect to the uninhibited control. The values depict the mean of three biological replicates ± standard error and are significant at P ≤ 0.001 (Tukey-HSD test; α = 0.05). The values with distinct letters are significantly different. (DAI: days after inoculation).

## Gene expression analysis:

Real-time PCR analysis revealed that out of 18 PPIase genes, only three cyclophilin (PoxCYP18, PoxCYP23 and PoxCYP41) and two FKBP genes (PoxFKBP12-2 and PoxFKBP52) were expressed at all stages of growth under both control and salt stress conditions in *P. oxalicum*. Whereas the expression of PoxCYP18 at 4 and 10 DAI increased significantly under salt stress, the transcript levels of PoxCYP23, PoxCYP41, PoxFKBP12-2 and PoxFKBP52 at all stages of growth decreased substantially.




Effect of salt stress (15% NaCl) on expression of cyclophilin and FKBP genes in the mycelia of *Penicillium oxalicum*. Changes in the transcript levels were estimated by Real-time PCR analysis using ACTIN gene as reference. The fold change presented is with respect to the control lacking NaCl. The values depict the mean of three biological replicates ± standard error and are significant at P ≤ 0.01 (Tukey-HSD test; α = 0.05). The values with distinct letters are significantly different. (DAI: days after inoculation).

## Conclusion

The halotolerant strain of *P. oxalicum* depicted significant enhancement in the mycelial PPIase activity in the presence of 15% NaCl, thus, highlighting the role of these enzymes in salt stress adaptation. The stress-induced increase in PPIase activity at 4 and 7 DAI in *P. oxalicum* was associated with higher expression of PoxCYP18. Characterization of PPIases in *Penicillium* spp. will provide an important database for understanding their cellular functions and might facilitate their applications in industrial processes through biotechnological interventions.

## Publication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